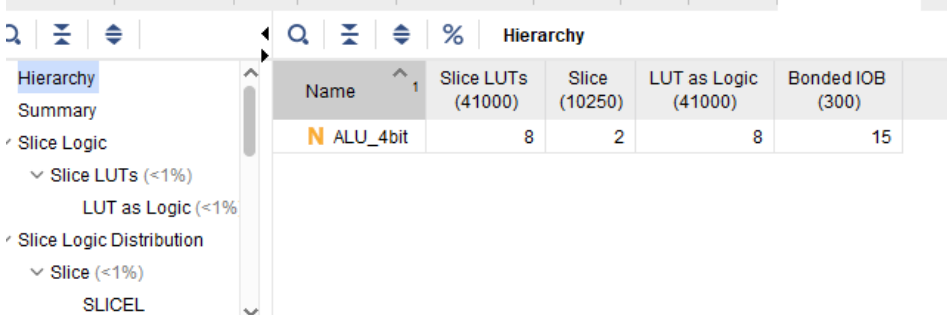
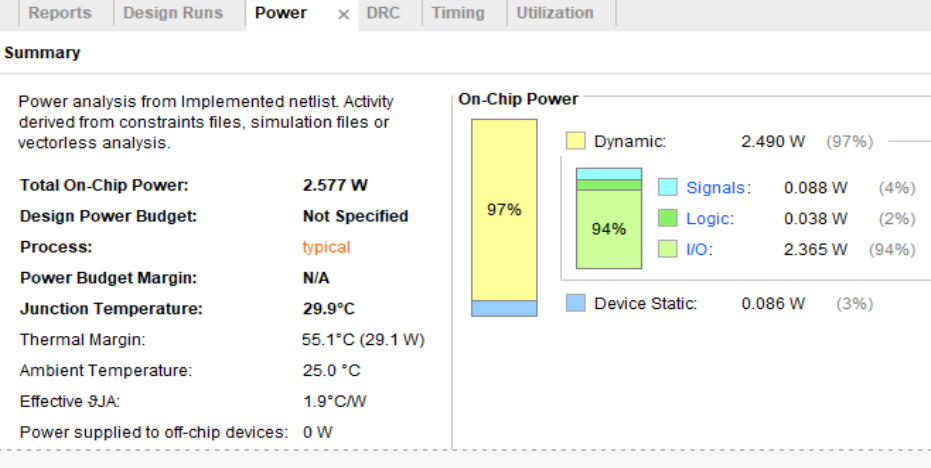
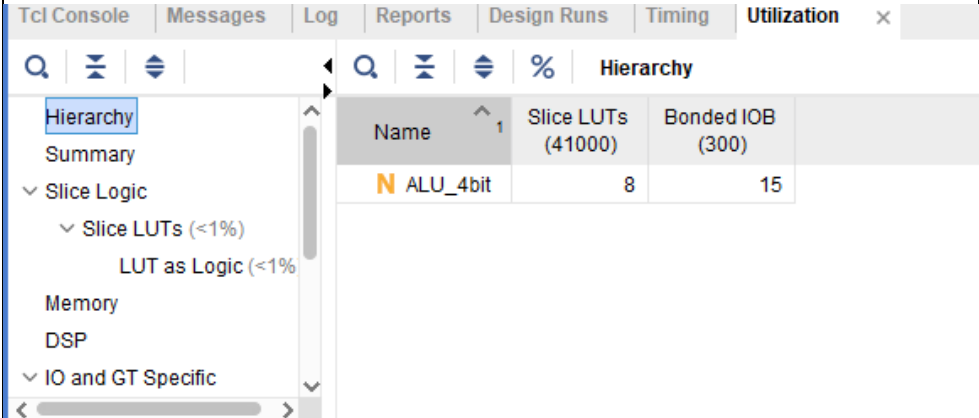
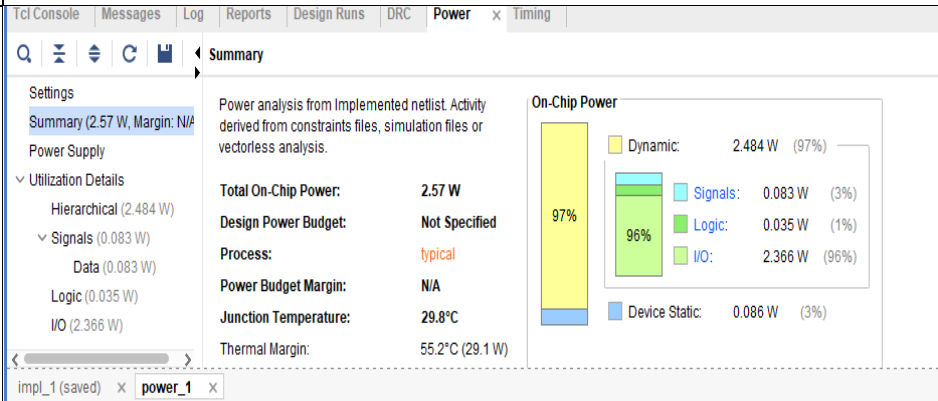
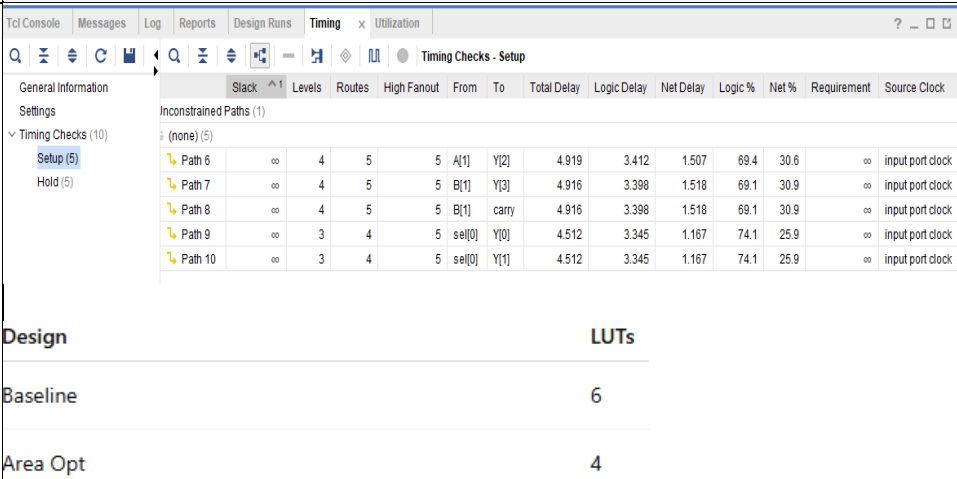


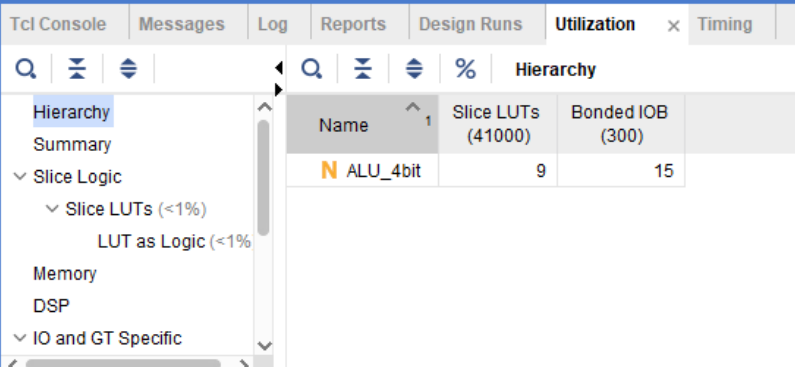
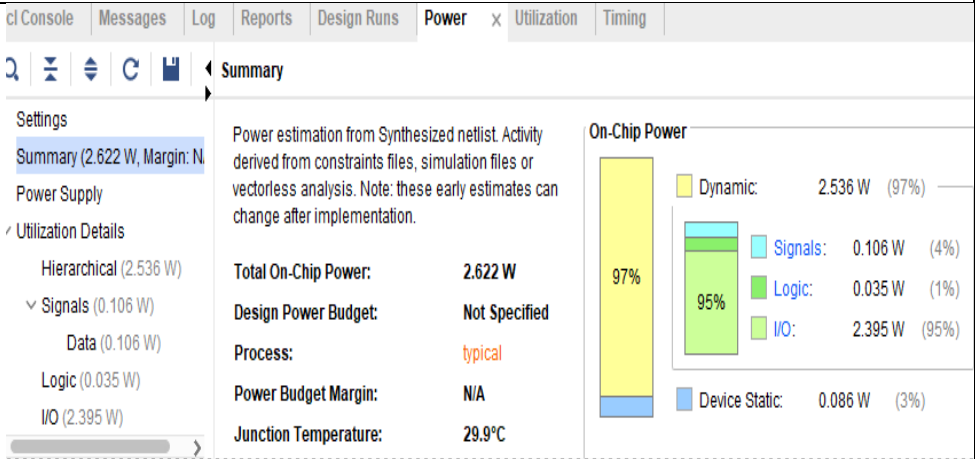
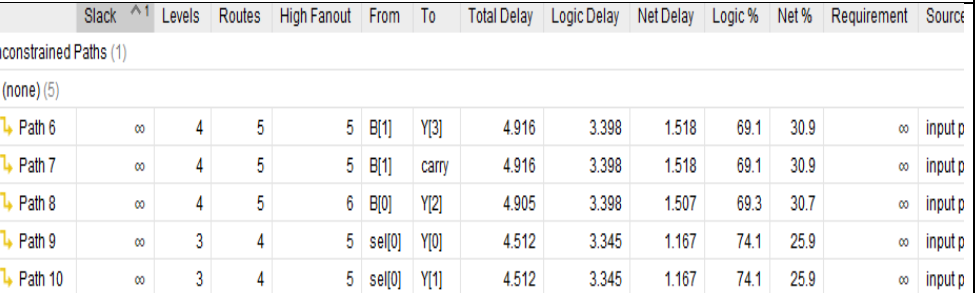
Area, Power, and Performance Trade-Off Analysis

Problem 1: Baseline Design																							
Aim	Design a 4-bit ALU supporting: - Addition - Subtraction - AND OR																						
LUT COUNT area																							
Power Consumption																							
Critical Path Delay	<table border="1"> <thead> <tr> <th>Component</th><th>Delay (ns)</th></tr> </thead> <tbody> <tr><td>IDELAYCTRL</td><td>0.00</td></tr> <tr><td>IBUFDS</td><td>0.00</td></tr> <tr><td>GTXE2_COMMON</td><td>0.00</td></tr> <tr><td>GTXE2_CHANNEL</td><td>0.00</td></tr> <tr><td>PHASER_OUT/PHASER_OUT_PHY</td><td>0.00</td></tr> <tr><td>PHASER_IN/PHASER_IN_PHY</td><td>0.00</td></tr> <tr><td>IDELAYE2/IDELAYE2_FINEDELAY</td><td>0.00</td></tr> <tr><td>ODELAYE2/ODELAYE2_FINEDELAY</td><td>0.00</td></tr> <tr><td>IBUFDS_GTE2</td><td>0.00</td></tr> <tr><td>ILOGIC</td><td>0.00</td></tr> </tbody> </table>	Component	Delay (ns)	IDELAYCTRL	0.00	IBUFDS	0.00	GTXE2_COMMON	0.00	GTXE2_CHANNEL	0.00	PHASER_OUT/PHASER_OUT_PHY	0.00	PHASER_IN/PHASER_IN_PHY	0.00	IDELAYE2/IDELAYE2_FINEDELAY	0.00	ODELAYE2/ODELAYE2_FINEDELAY	0.00	IBUFDS_GTE2	0.00	ILOGIC	0.00
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Result	The baseline ALU design shows balanced performance , with moderate area utilization, acceptable delay, and low power consumption. It serves as a reference for comparing area and performance optimized designs in later problems.																						

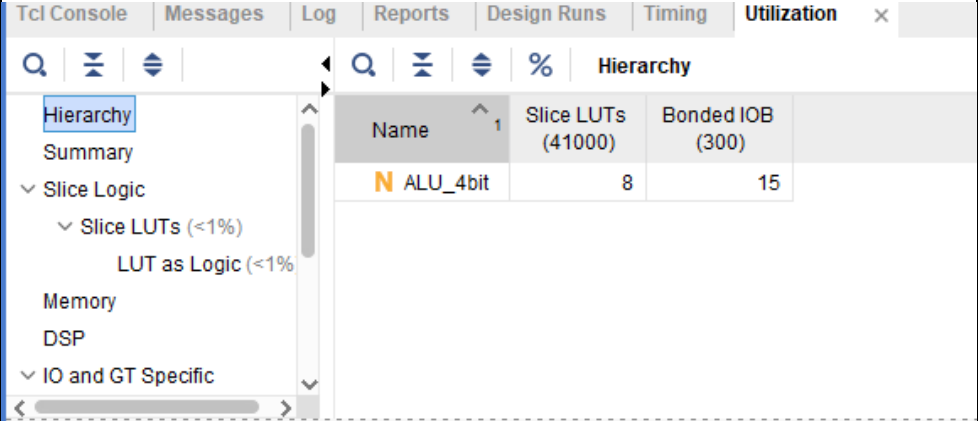
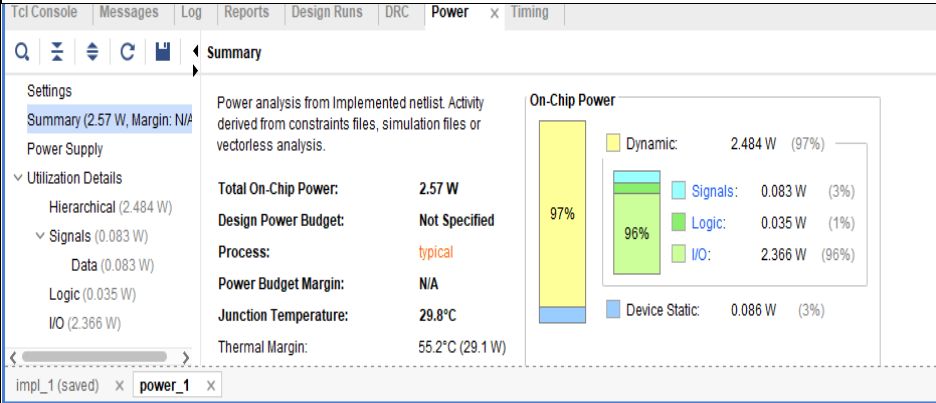
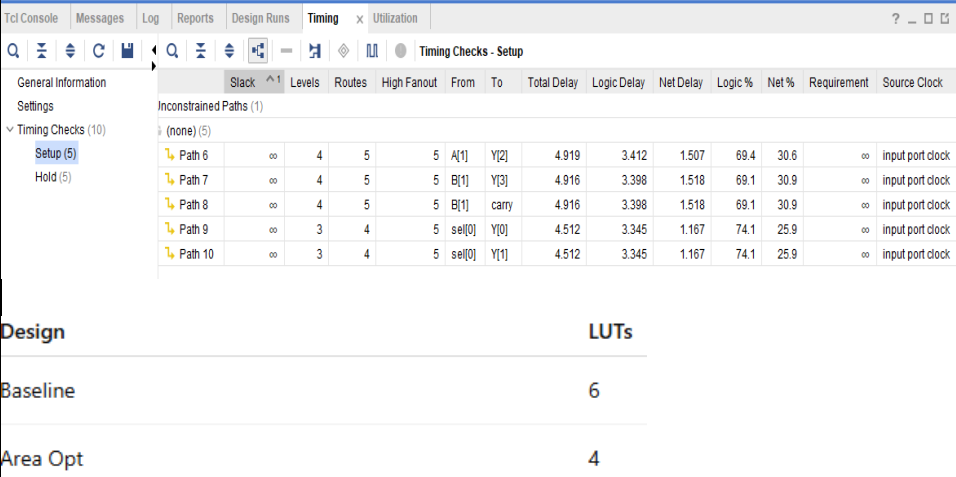
Problem 2: Area Optimization

Aim	Apply area optimization strategy in synthesis																																																																																																														
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Problem 3: Performance Optimization

Aim	Apply performance optimization strategy
LUT COUNT area	
Power Consumption	
Critical Path Delay	
Result	Performance optimization reduces critical path delay and improves speed (higher frequency), but increases LUTs, registers, and overall power consumption.

Problem 4 :Performance Optimization

Aim	Apply performance optimization strategy
Improvement in speed	
Power Consumption	
Critical Path Delay	
Result	Hence, Area optimization reduces LUTs and registers with slight power reduction, but increases delay and decreases the maximum operating frequency.